100 Days of Linux - Day 055

Title: Secure File Transfer with scp

Today's Goal
Learn how to securely copy files and directories between Linux systems using the **scp** command.

Why This Matters
Developers and data engineers frequently transfer scripts, datasets, configuration files, and backups between local computers and remote Linux servers.

Command of the Day: scp

Common Syntax:
scp file.txt user@server:/home/user/
scp user@server:/home/user/file.txt .
scp -r project/ user@server:/home/user/
scp -P 2222 file.txt user@server:/home/user/
scp -i ~/.ssh/id_ed25519 file.txt user@server:/home/user/

Common Options:
-r = Copy directories recursively
-P = Specify SSH port
-i = Use a specific SSH private key

Beginner Lesson
The **scp** command uses SSH to securely transfer files. You can copy files from your computer to a remote server or download files from a server back to your computer.

Practice Questions

1. Run 'scp' without arguments and read the help or usage message.

2. Write the command to copy report.txt from your computer to /home/ubuntu/ on a server at 192.168.1.100.

3. Write the command to download employees.csv from /home/ubuntu/ on the same server to your current directory.

4. Write the command to recursively copy a folder named project to the remote server using scp.

5. Display your current working directory after completing today's tasks.

Hints

Today's focus is learning the syntax. If you don't have a remote server, practice writing the commands correctly.

Mini Challenge

Create a folder named backup_project containing three sample files. Write the scp commands to upload the folder to a server and download it back to a folder named restored_project.

Common Mistakes
Forgetting the colon (:) after the remote hostname or using lowercase -p instead of uppercase -P for the port.

Did You Know?
Because scp uses SSH, all transferred data is encrypted while it travels across the network.

Pro Tip
Before using scp, verify that you can connect to the server successfully with ssh.

Answer Key (Instructor Only)
Q1: scp
Q2: scp report.txt ubuntu@192.168.1.100:/home/ubuntu/
Q3: scp ubuntu@192.168.1.100:/home/ubuntu/employees.csv .
Q4: scp -r project ubuntu@192.168.1.100:/home/ubuntu/
Q5: pwd

Homework
Practice writing scp commands for different files and folders until you can easily remember the direction of the copy.

Next Day Preview
Tomorrow you'll learn the ip command to inspect network interfaces, IP addresses, and routing information on Linux.